THE PERMANENT MAGNETISM OF THE MULL LAVAS

J. M. Bruckshaw and S. A. Vincenz

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Summary

An analysis of the natural magnetic polarization of the basalt flows of Mull shows that many are adversely magnetized but the mean direction of magnetization of each flow, based on observations of a number of specimens, is not always significant. For such flows measurements of the magnetic stability, response to heat treatment, etc., suggest that the original intensities have been modified by later events. Finally, the observed natural intensities could be acquired by cooling in a field similar in magnitude to that of the Earth's present field.

Introduction.—The detailed history of the Earth's magnetic field is only known from observations of the secular variation over a few centuries. Recently efforts have been made to extend this limited knowledge by a study of the natural residual magnetism of both sedimentary and igneous rocks. Thus the study of glacial varves (1) has suggested that the Earth's field has not changed significantly during very recent geological times. Similarly with igneous rocks, the examination of modern lavas has shown that the rocks acquire a residual magnetization in a direction corresponding to that of the Earth's field in which they cool (2), and observations on Quaternary Lavas (3) again indicate little change in the general direction of the Earth's field over the last million years, any small variation in direction comparing with the secular changes which have been observed directly. Many older igneous rocks, and also some older sedimentary rocks, show remarkable directions of magnetization, differing widely from the direction of the present field, and in most cases the direction of polarization is nearly opposite to that of the Earth's present field. Thus, a magnetic survey over the Pilandsberg Dyke System of South Africa (4), which is of age about 200×10^6 years, has yielded magnetic anomalies compatible with an adverse magnetization of the whole system. In many other places, e.g. Germany, Iceland, Japan, New Zealand, etc., other igneous rocks show similar characteristics. In particular, the suite of tholeiite dykes of the North of England (5) has been examined at a number of points over an area of some 200 miles by 80 miles. Here the investigations consisted of a determination of the magnetic anomaly in the vertical component of the Earth's field produced by the dykes at a number of sites, and at a selected number of sites oriented rock specimens were collected and their magnetic properties determined in the laboratory. These observations demonstrated that in general the whole of the system possessed adverse magnetization. It is significant here that the magnetic anomaly, predicted on the basis of the properties of the collected specimens, agreed closely with that observed, so that there could be no doubt of the representative nature of the specimens.

At present there is no universally accepted explanation of the adverse residual magnetization of these igneous rocks. At least two possibilities arise: in one the adverse magnetization is due to the fact that the ambient field in which the rocks cooled differed widely in direction from the present field, while in the other the origin of the abnormal directions lies in the intrinsic magnetic properties of the ferromagnetic constituents in the mineral assemblage of the rocks. To resolve the problem further data are necessary concerning the magnetic properties of the rocks themselves, of their mineral constituents, and also of the distribution in time and space of the rock formations possessing this abnormal property.

As the tholeiite dykes represent the last phase of igneous activity centred upon the Isle of Mull, it seemed desirable to obtain information concerning the magnetic state of the rocks intruded and extruded during earlier phases of the activity. The initiation of the activity (**6**) which occurred during the Tertiary Period consisted of the quiet extrusion over wide areas of basalt and mugearite lavas. The superimposed sequence of flows of the earlier lavas, the plateau lavas composed of an olivine-rich basalt, attained a thickness of many thousands of feet, and even today a thickness of some 3000 ft. remains. These were followed by the central lavas, having a much smaller olivine content, and which later took the form of a Kilauean dome with gentle flanks centred on a caldera. Later stages in the area took the form of a large-scale intrusion of cone-sheets and ring dykes about a centre which migrated in stages towards the north-west. The superimposed lava flows obviously present a time sequence of events, and because of the displacement of the centre of activity the order in which many of the later formations occurred can be determined. Thus the conditions permit the investigation of the change in the magnetic character of the rocks with time, although a time scale of years cannot be assigned to them. The final phase of the activity was characterized by the intrusion of numerous dykes, of which the tholeiite dykes form a part. The large-scale intrusion about the centres of activity has resulted in the pneumatolysis of the earlier lavas in their vicinity, a modification which is confined to a rough circle of about 8 miles radius around the centres.

The present paper deals with the examination of the lava flows only, and specimens were obtained from four sites, two within the zone of pneumatolysis and two outside it. From Site No. 1 (Fig. 1), in the Aros River valley, 42 oriented specimens were collected from five flows, and from Site No. 2 in Allt na Teangaidh valley a further 37 specimens from another five flows were taken. It is probable that the flows sampled at the latter site were not consecutive because of the paucity of exposures, but there is no doubt about their chronological sequence. The other two sites lay within the zone of pneumatolysis. Site No. 3 was situated on the shore of Loch na Keal some $3\frac{1}{2}$ miles along the coastal road leading south-west from Knock and they were sampled in sequence starting with the oldest on the shore of the Loch. Site No. 4 was to the north of the coastal road running along the north shore of Loch Beg. This last was a folded region, and microscopic examination of the specimens showed that many of the rocks had been crushed. It was impossible to distinguish the individual flows, but a total of 26 specimens were collected without reference to their sequence. Elsewhere the local geology revealed that no movement in excess of 10° had influenced the flows, and an examination of the specimens showed no evidence of crushing whatsoever.

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The method of observation.—Each flow was sampled throughout its thickness, the number of samples per flow ranging from 5 to 11, the average being 8. When sampling, the horizontal plane was marked on the specimen *in situ*, the top of the specimen was indicated together with an arrow giving the direction of the magnetic meridian. In a few cases this last mark was modified since it was found that some rocks were sufficiently magnetic to produce a marked deviation of a compass placed near them. A sight on a distant feature was then used to give the direction. From the oriented specimens cubes of 2 cm edge were cut in the laboratory, and in all cases one face of the cube was cut parallel to the horizontal plane *in situ* and another face was cut perpendicular to the magnetic meridian. The combined effects of the errors in the marking of the specimens and in cutting them were less than 2° in the horizontal plane and probably less than 4° in the direction of the magnetic meridian.

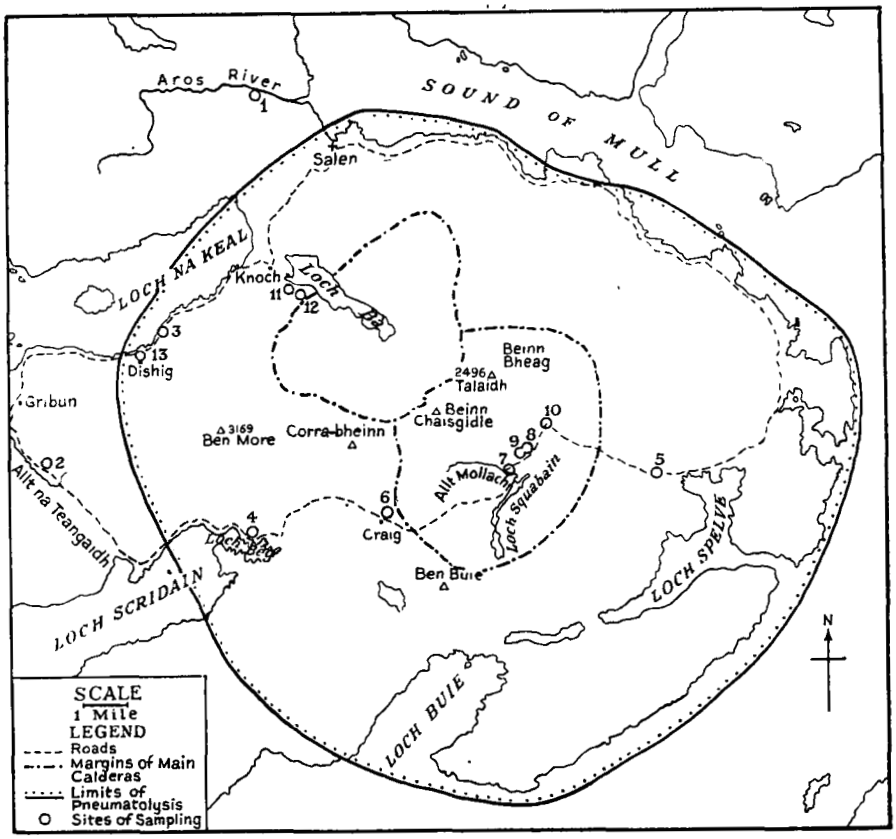


FIG. 1.—*The positions of the sampling sites.*

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Measurements were made to determine the intensity and direction of the permanent polarization of each cube and also the volume susceptibility. In the intensity observations the specimen was rotated at about 30 cycles per second near a coil system, the magnitude of the e.m.f. set up in the coils being a measure of the intensity and its phase being related to the direction. Measurements made with rotation about three mutually perpendicular axes allowed each component to be

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determined twice. The equipment employed was an improved version of that described by Bruckshaw and Robertson (7) and was based upon an original equipment due to Johnson and McNish (8). The susceptibilities of the rocks were measured in a magnetic field of intensity equal to that of the Earth's field, the method also being described by Bruckshaw and Robertson (7).

Factors influencing the permanent magnetism of igneous rocks.—Experience has shown that in the majority of igneous rock formations there is a variation in the direction of magnetization throughout their volume. Extreme examples of observed variations in direction are shown in Fig. 2. This is to be expected since a number of factors may influence the direction after the rock has acquired permanent magnetism. Thus such factors as movement during cooling and subsequent to cooling, mechanical shocks, mechanical crushing, metamorphism including thermal and hydrothermal alteration, weathering and other chemical actions, any instability of the original magnetism, and the influence of external magnetic fields may all tend to alter the direction of polarization. Many of these factors will tend to give a random distribution of directions. Movement during

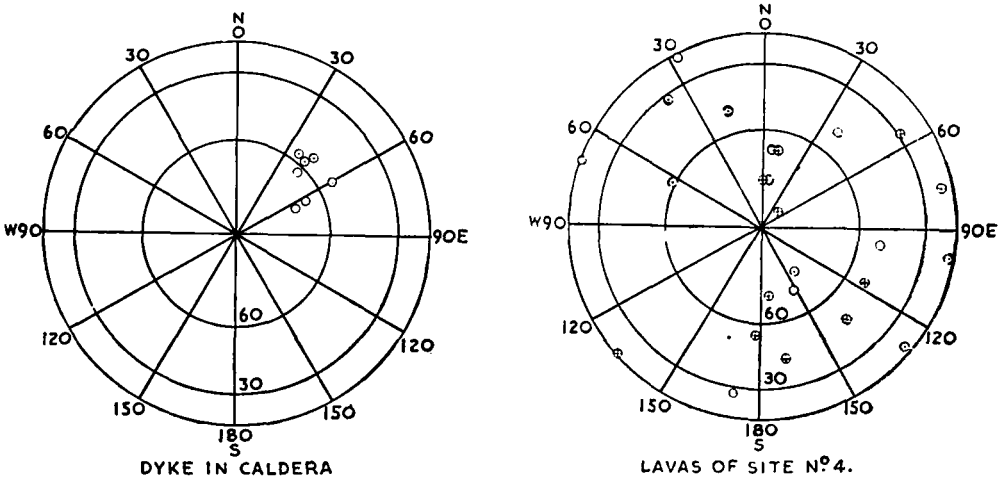


FIG. 2.—*Extreme examples of observed distributions in directions. Downward directions are shown by dots in circles and upward directions by crosses in circles.*

cooling of the type that may be associated with a lava flow, in which already solid and magnetized blocks may be twisted during further flow, will tend to destroy any initial uniformity of direction of the mass as a whole. Similarly, crushing by earth movement, or the influence of a number of lightning discharges, would tend to impose a random distribution of directions. Different factors, on the other hand, would tend to reduce or destroy the original intensity and possibly impose another direction. Weathering and other chemical actions would destroy the ferromagnetic constituents, and in the case of an adversely magnetized rock mechanical shocks, thermal effects, and an inherently unstable magnetic material would lead to a destruction of some of the original polarization and the superposition of a secondary direction controlled by the direction of the ambient field.

It is obvious that measurements on a single specimen would be insufficient to establish the direction of magnetization of a formation, and, when a number of specimens are examined, since they will show a scatter in the directions, some

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criterion is required to show if the mean direction is significant or not. The mean direction, if significant, will not necessarily be the original direction at the time of formation. An examination of the influence of the above factors shows that three possibilities arise. In the first the original intensity may be reduced without alteration of direction, or it may be destroyed. Secondly, the original intensity may be reduced or destroyed and another intensity, with a different direction, superimposed. Thirdly, a random distribution of directions may result. Only in the last case will any mean direction have no physical significance. The problem of assigning a significance or otherwise to the mean direction involves a determination of the probability of selecting a set of observed directions, associated with the specimens from a single formation, from a random distribution of directions. If this probability is low then considerable weight can be attached to the direction, but if the probability is high then the significance is small.

The statistical examination of the directions.—In the case of ν specimens from a uniformly magnetized formation, the sum of ν unit vectors with the observed directions will be ν. Any deviation from parallelism of the vectors will lead to a resultant less than ν, and in general the smaller this sum the greater the scatter in the individual directions.

The probability $W_\nu(r)\,dr$, that the sum of ν unit vectors, selected from a random distribution, lies between r and $r+dr$ has been examined by Rayleigh (**9**), Markoff (**10**), and Chandrasekhar (**11**), and it has been shown that

$$W_\nu(r) = \frac{1}{2\pi^2 r}\int_0^\infty \sin\rho r \left(\frac{\sin\rho}{\rho}\right)^\nu \rho\, d\rho.$$

When ν is large this approximates to the exponential form

$$W_\nu(r) = \{\exp(-3r^2/2\nu)\}/(2\pi\nu/3)^{3/2},$$

and, when ν is small, as in the present investigation,

$$W_\nu(r) = f(r)/2^{\nu+2}\pi r,$$

where $f(r)$ is a polynomial of degree $(\nu-2)$ with real coefficients. The probability $p_\nu(r)\,dr$ that the magnitude of the sum of the vectors lies between r and $r+dr$ is obviously

$$p_\nu(r)\,dr = 4\pi r^2\, W_\nu(r)\,dr,$$

for which it can be shown that

$$\int_0^\nu p_\nu(r)\,dr = 1.$$

This latter probability function $p_\nu(r)$ has been used in the present investigation. This function is zero for $r=0$ and $r=\nu$, and has a single maximum of the order of $\nu^{-1/2}$ at about $r=(2\nu/3)^{1/2}$. Thus, for large values of ν the maximum value of the probability function is itself small, and it was found convenient to compare the probability function for any set of observed directions with the maximum value for the number of specimens in the set.

The results of the measurements.—It is obvious that in determining the mean direction of a set of observed directions no account should be taken of the absolute intensities of magnetization, since these merely reflect the fortuitous quantity of the ferromagnetic constituents and their properties. This is of course inherent in the statistical method sketched above. Thus if ϕ_i and I_i are

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the observed azimuth (measured from magnetic north) and dip for an individual specimen, the sum for a group of unit vectors with the observed direction is

$$r=(x^2+y^2+z^2)^{1/2},$$

and the mean azimuth $\bar{\phi}$ and dip $\bar{I}$ are given by

$$\tan\bar{\phi}=y/x \quad \text{and} \quad \tan\bar{I}=z/(x^2+y^2)^{1/2},$$

where

$$x=\sum_1^\nu \cos\phi_i\cos I_i,$$

$$y=\sum_1^\nu \sin\phi_i\cos I_i,$$

and

$$z=\sum_1^\nu \sin I_i,$$

i.e. the sum of the components of the unit vectors in three mutually perpendicular directions. In the present work the x-axis was directed towards the north, the y-axis to the east and the z-axis downwards, and the usual convention of a positive downward dip was taken.

The results of the measurements on the flows are summarized in Table I. In this the angle δ is the computed angle between the mean direction and the direction of the present magnetic field, and in the penultimate column the maximum of the probability function for the value of ν concerned has been quoted.

TABLE I

Site No.	Flow No.	ν	$\bar{\phi}$	$\bar{I}$	δ	r	$p^\nu(r)$	$p_\nu(r)_{max}$	Q_n
	1	8	147°	−42°	147°	4·69	0·062	0·357	4·4 ±1·8
	2	11	156	−78	170	8·73	0·0001	0·310	3·8 ±1·5
1	3	8	105	−55	145	2·28	0·355	0·357	61·4 ±50
	4	8	17	−62	133	7·62	0·0001	0·357	2·4 ±1·1
	5	7	− 8	−40	111	3·10	0·293	0·384	8·6 ±6·1
	1	7	60	−80	154	4·70	0·043	0·384	3·5 ±1·0
	2	8	178	−71	176	6·33	0·0014	0·357	3·9 ±2·1
2	3	7	153	−69	171	2·92	0·325	0·384	3·6 ±3·0
	4	7	102	− 6	100	2·38	0·383	0·384	0·56±0·76
	5	8	− 96	−51	139	3·01	0·315	0·357	7·7 ±4·0
	1	6	165	−51	159	2·70	0·353	0·421	2·7 ±0·7
	2	8	−160	−67	172	5·72	0·007	0·357	7·9 ±8·8
3	3	5	174	−54	163	4·79	0·0015	0·460	3·1 ±1·6
	4	8	−155	−59	165	5·69	0·009	0·357	2·5 ±1·8
	5	7	−113	−16	113	3·28	0·255	0·384	13·4 ±26
	6	8	26	− 4	76	2·29	0·355	0·357	33·1 ±24
4	...	26	72	35	52	2·89	0·171	0·199	2·9 ±7·7

A comparison of the values of the probability function for the observed value of ν with the maximum value suggests that little significance can be attached to some of the mean directions. Thus, the directions observed for Site No. 1 flows 3 and 5, Site No. 2 flows 3, 4, and 5, and Site No. 3 flows 1, 5 and 6, could have been selected readily from a random distribution. On the other hand, the remaining eight give directions which appear to be significant. For these, all the dip angles are negative, and the majority of the declinations are in the second and

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third quadrants, nearly in opposition to the present declination. In all flows with a significant direction the angle δ exceeds 133°, so that the directions are approximately in opposition to the present field. Even those flows with a mean direction to which little significance can be attached, with one exception in flow 6 of Site No. 3, give values of δ in excess of 90°. Site No. 4 is of considerable interest since it reveals a close approximation to a random distribution (Fig. 2 (*b*)), a feature which can be attributed to the crushing shown by the rocks and associated with obvious folding at the site.

Since on any one site the flows can be taken as representing events over a few thousand years, the mean direction for the set of flows is of interest. These are given in the first three rows of Table II, the figures being computed on the basis of equal weights for the flows. A corresponding analysis is also given for the flows with significant directions ("good" flows), those whose directions have little significance and for all the flows. The specimens are treated in the same way, the results being equivalent to weighting the mean direction associated with any flow by an amount proportional to the value of r for the flow. As a result of this analysis there can be no doubt that the direction of magnetization of the flows examined is on an average opposed to that of the present field.

TABLE II

	No.	$\bar{\phi}$	$\bar{I}$	δ	r	$p_v(r)$	$p_v(r)_{max}$
Site No. 1	5	77°	−76°	154°	4·13	0·027	0·460
No. 2	5	141	−74	168	3·90	0·051	0·460
No. 3	6	−159	−67	172	4·04	0·075	0·421
"Bad" flows	8	96	−81	159	4·45	0·090	0·357
"Good" flows	8	166	−73	175	7·35	7×10^{-5}	0·357
All flows	16	151	−78	169	11·70	$2{\cdot}4\times10^{-5}$	0·250
"Bad" specimens	58	−139	−89	161	13·4	0·015	0·133
"Good" specimens	63	−174	−77	174	44·5	5×10^{-20}	0·128
All specimens	121	−173	−79	171	58·0	7×10^{-18}	0·090

In addition to the direction and intensity of the natural residual magnetism, the volume susceptibility K_n of each specimen was measured, and from it the ratio of the natural residual intensity J_n to the induced intensity K_nF was calculated, where F is the total field at the site. This ratio Q_n ($=J_n/K_nF$) is the natural Koenigsberger ratio, and for any one flow it showed a considerable variation. In the last column of Table I the mean value and the standard deviation are quoted. An examination of the values associated with the good flows shows that, with the exception of flow 2 of Site No. 3, the mean values lie between 2·4 and 4·4, and the standard deviation is about half the mean value. The remainder of the results are extremely varied; in some cases the value of Q_n is high (Site No. 1 flows 3, 5, and Site No. 3 flows 5, 6) and in other cases the value is abnormally small (Site No. 2 flow 4). The other flows with mean deviations of little significance do not show any remarkable values of Q_n although it might be claimed that the standard deviation is either large or small. It is worth noting that the value of the susceptibility did not change by a factor greater than 3 and the wide range of the Q_n values was mainly due to the value of J_n which, in one case, ranged from 7×10^{-3} up to 111×10^{-3} c.g.s. units.

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Laboratory investigations

(*a*) *Stability.*—In 1938, Thellier (**12**) found a marked instability in the natural residual magnetism of certain rocks, including some from the Isle of Mull. To investigate this, a number of typical samples were stored with their x-axes towards the north so that an adversely magnetized specimen was in a demagnetizing field, and measurements of intensity were made at intervals of a few months. The majority showed small changes which could be attributed to experimental error. A few showed a greater decrease, and further tests were made on these and on a few stable specimens. For a period of ten days the specimens were stored with their x-axes towards the north, then to the south, and finally towards the north again. Nearly all specimens showed a change of about 1 per cent in intensity but some showed a greater variation. By far the largest change was exhibited by specimens from Site No. 2 flow 4, for which a 70 per cent change in intensity occurred. This may well account for the very low value of Q_n. It is to be noted that the mean direction of this flow has little significance and in all probability does not refer to the direction of original magnetization. For three specimens from Site No. 2 flow 3, also a "bad" flow, changes in intensity of 4 per cent, 12·5 per cent and 25 per cent were observed, and flow 5 of Site No. 3 exhibited a 10 per cent change. All these large changes were associated with flows of small significance of the mean direction, but all such flows did not necessarily exhibit such instability. In some of the flows with significant directions changes of 2–3 per cent occurred occasionally, in particular flow 4 of Site No. 1 and flow 2 of Site No. 2.

This instability is possibly due to the presence in the rock of at least two constituents, one having a low coercivity and responding to the ambient field. If high coercivity material predominated, the intensity impressed at normal temperatures by the Earth's field would be negligible, and the observed direction would approximate to the original direction. With a predominance of low coercivity material an unstable rock would result and one in which the observed direction would have no relation to the original. If the intensities arising from the two constituents were comparable, then a formation similar to flow 4 of Site No. 2, with a low value of Q_n, might well result.

(*b*) *Influence of heating.*—Within the zone of pneumatolysis, at Site No. 3, the temperature may have increased to 200 deg. C after the original cooling of the lava. Inspection of Table I does not suggest any marked influence of this reheating, but tests were made to examine any possible effects. It has been shown by Nagata (**13**) and Thellier (**14**) that the thermo-remanence acquired by a rock in cooling in a weak field between temperatures θ_2 and θ_1, is given by

$$J_{\theta_1}^{\theta_2} = \int_{\theta_1}^{\theta_2} H\,P(\theta)\,d\theta, \qquad (1)$$

where H is the field in which the rock cools and $P(\theta)$ is a characteristic of the rock. Thus if the field in which the rock cooled initially from above its Curie point was F_1, and a subsequent reheating and cooling took place in a field F_2, then

$$J_n = \int_{t_0}^{\theta_c} F_1\,P(\theta)\,d\theta$$

for the rock as found and

$$J_t = \int_{t}^{\theta c} F_1\,P(\theta)\,d\theta + \int_{t_0}^{t} F_2\,P(\theta)\,d\theta$$

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after heat treatment, where t is the temperature of reheating t_0, is the normal temperature and θ_c is the maximum Curie temperature of the rock constituents. Part of the original magnetization is destroyed and a new polarization added.

The attitude of the specimens during the heat treatment was as found *in situ*. The majority of the specimens showed a reduction in the intensity of magnetization and a number exhibited a definite but small rotation of the direction. All adversely polarized rocks, however, remained so after the heat treatment, and there was no significant difference between the rocks from the two regions. It would appear that with the lavas examined the thermal effects associated with pneumatolysis would not have altered in a material manner the results which have been obtained.

The observed decrease in the intensity varied from 10 up to 50 per cent, but in all cases there was an increase in the susceptibility up to 10 per cent. The change in intensity agrees with the above equations since adverse magnetization would be destroyed and normal magnetization imposed in its place. On the assumption that $F_1 = F_2$, i.e. the magnitudes of the part destroyed and added were equal, it was possible to test roughly the vector addition by calculating the dip of the Earth's field in the laboratory. The values so computed varied from 22° up to 63°, and in view of the assumptions made and the observed changes in the susceptibilities these can be regarded as satisfactory. Indeed, with an improved technique it might be possible to deduce the ratio of F_1/F_2. As there was no significant difference between the results using samples inside and outside the zone of pneumatolysis, it suggests that the field acting at the time of any second cooling was similar to that during the original cooling.

The increase of susceptibility may be due to the production of new ferromagnetic materials by the decomposition of the ferro-magnesian minerals or to changes in the lattice of the existing magnetic constituents. In view of the low temperatures involved the latter seems more probable. A number of the specimens were heated to 200 deg. C a second time but no further changes were observed. In two cases the heat treatment produced very marked changes. One specimen from the site of the crushed rocks gave a very large change in direction while the second, from Site No. 2 flow 4, was rendered so unstable that the residual magnetism could not be measured, thus matching the instability already discussed. In a matter of 70 minutes the intensity of one component decreased by more than 20 per cent and the direction changed by 26°. In general such abnormal behaviour was always associated with formations which appeared to be magnetized at random.

(*c*) *Measurement of thermo-remanence.*—Finally, experiments were carried out to ensure that the residual magnetism was thermo-remanent in origin. The specimens were heated to a temperature above the Curie point of their constituents and allowed to cool in the Earth's field. The thermo-remanent intensity J_c, which always had the direction and sense of the magnetizing field, and the susceptibility K_c were determined after treatment, and the ratio $Q_c = J_c/K_c F$ computed, where F denotes the field in which the rocks cooled. It follows from equation (1) that J_c is proportional to the field if this is constant, and so also is the induced intensity at room temperatures. Thus the ratio Q_c is independent of the field and is a characteristic of the rock. Thus if F_1 is the field in which the rock cooled originally and acquired the observed natural intensity J_n, then

$$Q_c = J_n/K_n F_1 = Q_n F/F_1,$$

so

$$F/F_1 = Q_n/Q_c = T.$$

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Thus a comparison of the values of Q_n and Q_c should allow a comparison of the present field and that in which the rock cooled initially. A number of assumptions, which cannot be fully justified, are involved. No allowance has been made for a reduction in intensity by factors operating subsequent to cooling. Certain instabilities have been noted in the laboratory and even reheating to 200 deg. C may modify the system. In addition, it is not clear that the magnetic properties of the rocks after the laboratory heating are the same as those before. Quite large changes in the susceptibilities were recorded. Thus, for site No. 1 there was an average decrease to 0·69 of the susceptibility before heating, although one sample gave an increase of 40 per cent. On Site No. 2 a similar but smaller average decrease occurred, but on Site No. 3, within the zone of pneumatolysis, an average increase of 24 per cent was noted. Thus no great accuracy can be claimed but it may be anticipated that the field may be of the right order.

For the three sites in order the values of Q_c were $14{\cdot}3 \pm 2{\cdot}5$, $12{\cdot}4 \pm 2{\cdot}7$, and $10{\cdot}9 \pm 3{\cdot}5$ respectively, those from the zone of pneumatolysis having the smallest value and the greatest scatter. The corresponding values of T were $3{\cdot}8 \pm 3{\cdot}7$, $4{\cdot}6 \pm 2{\cdot}3$, and $3{\cdot}8 \pm 1{\cdot}1$. Within each group a number of rocks, although having a value of Q_c comparable with the others, had an abnormally low value of T. From Site No. 1 three values were 0·57, 0·26 and 0·77 implying a low value of Q_n, and all specimens were from formations having high values of $p_v(r)$. One specimen had the high value of 27 for T, and again was from a formation which appeared magnetized at random. In general it would seem that the intensities of natural magnetization are compatible with thermo-remanent magnetization acquired in a magnetic field of strength similar to that existing today. This conclusion is strengthened by the results of an experiment in which a demagnetized rock was exposed to magnetizing fields of different intensities and observing the acquired intensity of magnetization. In this way it was found that fields ranging between 3·2 and 62 gauss were necessary to impress the observed natural intensities, the lower figure being obtained with a specimen which had exhibited poor stability. With this mode of magnetization very large and adverse fields would be necessary to yield the observed natural intensities.

Discussion.—There can be little doubt that many of the basalt flows of Mull are adversely magnetized, the polarization being thermo-remanent in origin. As the last occasion when the rock temperatures were above their Curie point was at the time of their extrusion the intensities appear contemporary with this, and the existence today of adverse magnetization demonstrates the extraordinary stability of thermo-remanent magnetism. Laboratory tests have revealed certain cases of instability, but for such specimens there is reason to believe that the formations have been exposed to later influences which have modified their magnetic properties. Certainly, the mean direction of magnetization for such formations, determined on the basis of a number of samples, is not significant. Certain formations however, which appear stable under laboratory conditions, also show a tendency to random magnetization.

The results do not yield any direct evidence of the conditions under which the adverse thermo-remanent polarization was acquired. Nevertheless, on the simple assumption of an ambient adverse magnetic field, the observed intensities of magnetization could have been impressed by a field comparable in intensity with that of the present field. The normal ferromagnetic properties exhibited by the rocks eliminate the possibility of a constituent similar to that separated by

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Nagata (**15**) from a dacite and which exhibits adverse thermo-remanent magnetism in the laboratory. Again, the two-component mechanism advanced by Néel (**16**) would not appear promising because of the normal thermo-remanence exhibited by the rocks on heating. Graham (**17**) however has suggested that the laboratory heat treatment may destroy the original magnetic properties, and if this is so the possibility of a direct check on the adverse nature or otherwise of the original field is removed. Hospers (**18**) has pointed out that on the basis of the two-component mechanism there should be a reversal in the direction of the intensity of magnetization as the rock is heated. This reversal should occur just below the lower Curie temperature of the two components. Thus, by heating the rock in zero field, any radical change in the magnetic properties of the ferromagnetic constituents as postulated by Graham should be accompanied by a loss of all or part of the original intensity of magnetization at whatever temperature the change occurs or, if Néel's theory is applicable, there should be a reversal in the direction of magnetization. These investigations are now in progress and they may well throw new light on this problem.

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Imperial College of Science,
London, S.W.7:
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References

(1) McNish, A. G. and Johnson, E. A., *Terr. Mag. Atmos. Elect.*, **43**, 4, 1938.
(2) Chevallier, R., *Ann. de Physique*, **4**, 5, 1925.
(3) Hospers, J., *Nature, Lond.*, **168**, 1111, 1951.
(4) Gelletich, H., *Beitr. Geophys.*, **6**, 337, 1937.
(5) Bruckshaw, J. M. and Robertson, E. I., *M.N., Geophys. Suppl.*, **5**, 308, 1949.
(6) Tertiary and Post-Tertiary Geology of Mull, *Mem. Geol. Surv. Scotland*, 1924.
(7) Bruckshaw, J. M., and Robertson, E. I., *J. Sci. Inst.*, **25**, 444, 1948.
(8) McNish, A. G. and Johnson, E. A., *Terr. Mag. Atmos. Elect.*, **43**, 393, 1938.
(9) Lord Rayleigh, *Phil. Mag.*, **40**, 673, 1880 and **47**, 246, 1889.
(10) Markoff, A. A., *Wahrscheinlichkeitsrechnung*, 16 and 33, Leipzig, 1912.
(11) Chandrasekhar, S., *Rev. Mod. Phys.*, **15**, 1, 1943.
(12) Thellier, E., *Ann. Inst. Phys. Globe*, **16**, 157, 1938.
(13) Nagata, T., *Earthquake Res. Inst. Bull.*, **21**, 1, 1943.
(14) Thellier, E., *C. R. Acad. Sci., Paris*, **223**, 319, 1946.
(15) Nagata, T., *Nature, Lond.*, **169**, 704, 1952.
(16) Néel, L., *Ann. de Géophys.*, **7**, 90, 1951.
(17) Graham, J. W., *Journ. Geophys. Res.*, **57**, 429, 1952 and **58**, 243, 1953.
(18) Bruckshaw, J. M., *Nature, Lond.*, **171**, 500, 1953.

1365246x, 1954, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1954.tb03041.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License